

Introduction to machine learning

Machine Learning

Introduction to machine learning

“[Machine Learning is the] field of study that gives computers the ability to learn without being explicitly programmed.” Arthur Samuel 1959:

Introduction to machine learning

What is machine learning?

1. Is a process of enabling a computer based system to learn to do tasks based on well defined statistical and mathematical methods
2. The ability to do the tasks come from the underlying model which is the result of the learning process. Sometimes the ability comes from an mathematical algorithm
3. The model generated represents behaviour of the processes that were earlier performed before machine learning
4. The model is generated from huge volume of data, huge both in breadth and depth reflecting the real world in which the processes are performed
5. The more representative data is of the real world, the better the model would be. The challenge is how to make it a true representative

Introduction to machine learning

What do machine learning algorithms do?

1. Search through data to look for patterns
2. Patterns in form of trends, cycles, associations, classes etc.
3. Express these patterns as mathematical structures such as probability equations or polynomial equations

Introduction to machine learning

When is machine learning useful ?

1. Cannot express our knowledge about patterns as a program. For e.g. Character recognition or natural language processing
2. Do not have an algorithm to identify a pattern of interest. For e.g. In spam mail detection
3. Too complex and dynamic. For e.g. Weather forecasting
4. Too many permutations and combinations possible. For e.g. Genetic code mapping
5. No prior experience or knowledge. For e.g. Mars rover
6. Patterns hidden in humongous data. For e.g. Recommendation system

Introduction to machine learning

Where are machine learning based systems used (examples only)

1. Fraud detection
2. Sentiment analysis
3. Credit risk management
4. Prediction of equipment failures
5. New pricing models / strategies
6. Network intrusion detection
7. Pattern and image recognition
8. Email spam filtering

Introduction to machine learning

Machine Learning & Data Science

1. Machine learning is part of a larger discipline called Data Science
2. Data science is the process of applying science and domain expertise to data to extract useful information from data.
3. It includes application of all the statistical and mathematical tools and techniques to glean out the useful information from data using machine learning

Introduction to machine learning

Machine Learning Pre-requisites

1. Rich set of data representing the real world
2. Knowledge and skills in
 - a. Maths and statistics
 - b. Programming (Python, R, Java, Go)
 - c. Tools / frameworks such as Keras / TensorFlow
 - d. Domain knowledge